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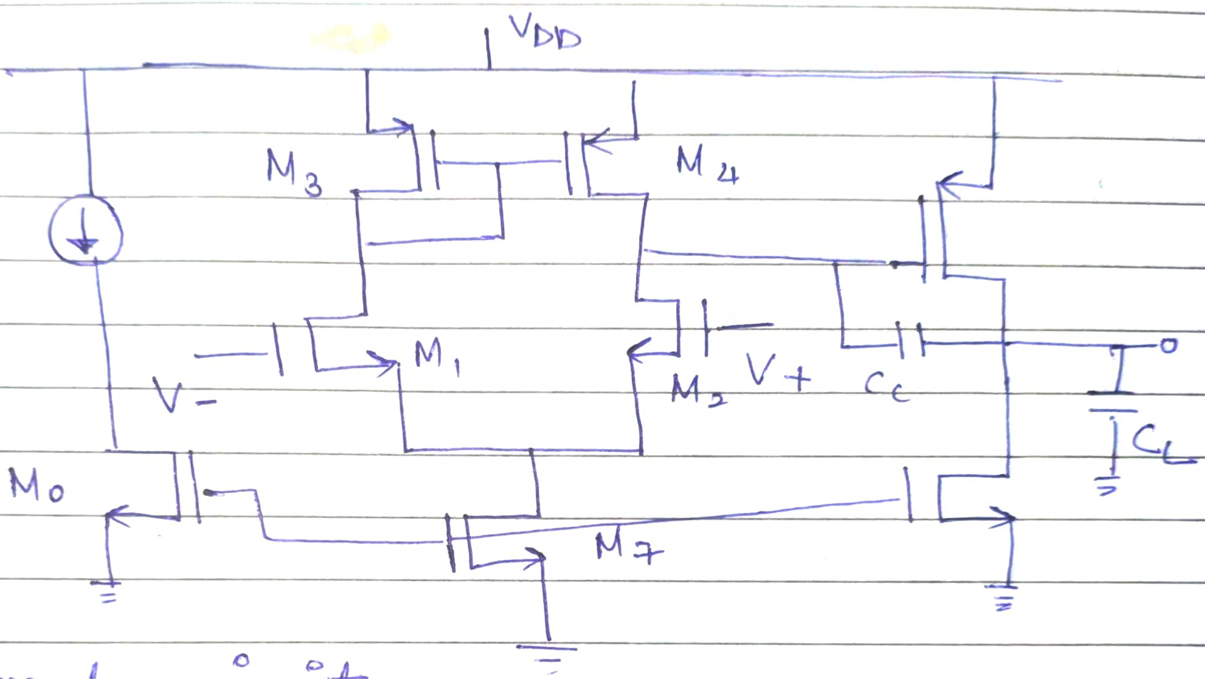
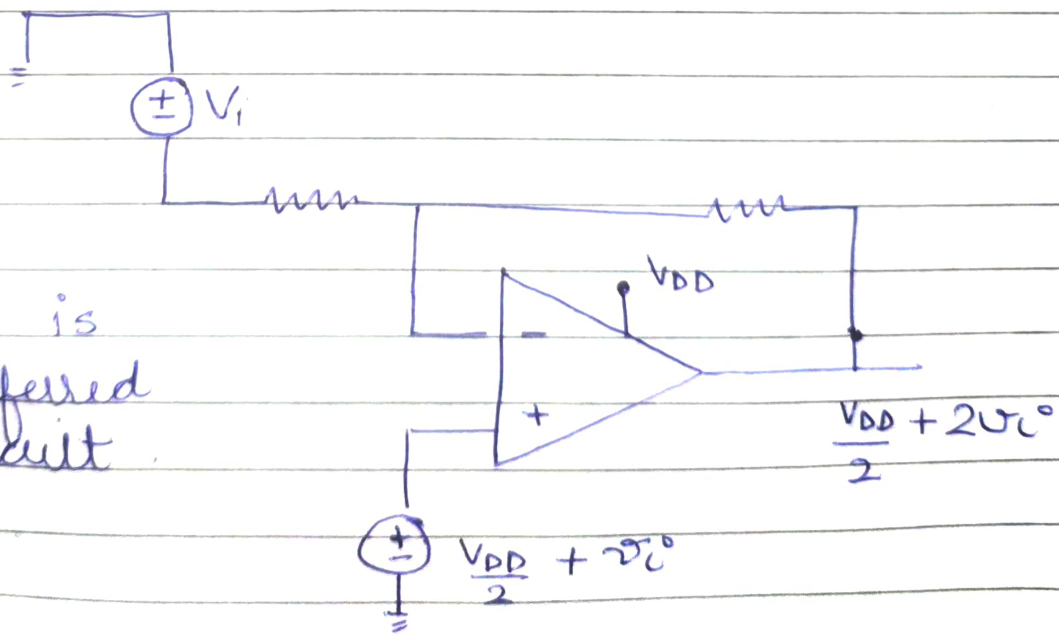
Project Report

Eklavya Jain
240102103

Circuit Diagram



Prefered circuit



This is prefered circuit.

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Specifications

i) atleast 40 dB (taking 60 dB) of DC Gain.
ii) GBW = 5 MHz.
iii) PM $\geq 60°$
iv) Slew rate $\geq$ 10V/µsec
v) ICMR (+1) = 1.6V   ICMR (−1) = 0.8V
vi) Let's take C_L = 10 pF
vii) Vdd = 1.8V
viii) $\mu_n C_{ox} = 200\ \mu A/V^2$ (initially)
$\mu_p C_{ox} = 100\ \mu A/V^2$ (initial)

Calculations

→ Slew rate = $\frac{I_7}{C_C}$ and for $\geq 60°$ phase margin.

$C_C = 0.4 C_L = 4$ pF $I_7 \geq 40 \mu A$

• M_1 & M_2

→ g_{m1} = GBW . Cc . 2π = 40π µS = 125 µS

→ $\left(\frac{W}{L}\right)_1 = \frac{g_{m1}^2}{\mu_n C_{ox} I_{D7}} = 1.69$

→ $\text{||ly} \left(\frac{W}{L}\right)_2 = 1.69$

→ $\sqrt{\frac{I_{D5}}{\beta_3}} = V_{DD} - |V_{th_3}| + V_{Th_1} - V_{in_{max}}$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{I_5}{\mu_p C_{ox} \left(V_{DD} - |V_{th_3}| + V_{Th_1} - V_{in\,max}\right)^2}$$

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$$\left(\frac{W}{L}\right)_{3,4} = 20.40$$

Since $I_7 \geq 40\,\mu A$ and $I_{ref} = 4$ so 10 times current. So, factor is ≥ 10.
Assume $\left(\frac{W}{L}\right)_{M_0} = 2$. then we must have $\left(\frac{W}{L}\right)_7 \geq 2 \times 10 = 20$.

I took it be 40 for good gain for reasonable phase margin ($\sim 60°$).

$$g_{m5} = 2.2\, g_{m4} \left(\frac{C_L}{C_c}\right)$$
$$\& \quad g_{m5} >> 10 g_{m1}$$

$$V_{SG_4} = V_{SG} = 5$$
$$\left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_4 \left(\frac{g_{m5}}{g_{m4}}\right)$$

$$\left(P_2 = \frac{g_{m5}}{2\pi C_L}\right)$$ (Comes from 2nd stage).

$\therefore$ $P_2 \geq 11\,MHz$ &

$$11 \times 10^6 = \frac{g_{m5}}{2\pi C_L}$$

$$g_{m5} \geq 697\,\mu s$$

Now since overdrive in both are same ∴ $V_{SG} - |V_{TP}|$ is same both M_4 and M_5.

$$g_{m4} = \sqrt{2 u_p C_{ox} \left(\frac{W}{L}\right)_4 I_{D4}} = 2861\ \mu s$$

$$\Delta V_{04} = \frac{2 I_{D4}}{g_{m4}} - 0.14V$$

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So for better phase margin we already got

$g_{m5} \geq 6911 \, \mu s$

we must have $\Delta V_5 = \Delta V_6 = 0.14V$

$$g_{m5} = \frac{2I_{D5}}{V_{ov5}}$$

$I_{D5} = 48.37$ (mu A)

Let's assume ($\approx 55 \mu A$)

$$\left(\frac{W}{L}\right)_5 = 56.2375$$

$W = 25.551$ um $\quad L = 0.45 \mu m$

now M_6 must be able to pull this current off

$\left(\frac{55}{4}\right)$ = multiplier

$$\frac{55}{4} \times 1.69 \approx \frac{55}{4} \times 2 = 27.5 = \left(\frac{W}{L}\right)_6$$

Let's check what we get from LT spice :

i) dc gain :- 64.783 dB (≥ 60dB)

ii) P.M at 0 dB $= 65°$.

iii) GBW = 4.7 MHz

iv) Currents in each transistor

All are dc currents.

$M_0 = 4 \mu A$ $\quad M_2 = 27.74 \mu A$ $\quad M_4 = 27.74 \mu A$

$M_1 = 27.87 \mu A$ $\quad M_3 = 27.87 \mu A$ $\quad M_7 = 55.62 \mu A$

$M_5 = 46.8984 \mu A$ $\quad M_6 = 46.8984 \mu A$.

now $r_0 = \frac{1}{\lambda I_D}$ & $g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}$

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r_o

Tr No	r_o (KΩ)	gm (uS)	ΔV	W/L	I_D (uA)	W	L (um)
0	2500	60.7	0.132	2.0	4	1um	0.5
1	360	147	0.378	1.69	27.8	0.85um	0.5
2	360	147	0.378	1.69	27.8	0.85um	0.5
3	360	337.2	0.165	20.45	27.8	10.22 um	0.5
4	360	337.2	0.165	20.45	27.8 ~~46.9~~	10.22 um	0.5
5	180	1011.6	0.110	40	46.9	[illegible] um	0.5
6	213	692.3	0.135	51.1	46.9	12.5 um	0.5
7	213	734.4	0.128	25.5	55.5	20um	0.5

GBW

PM ≈ 65°

Cursor 1

V(n004)

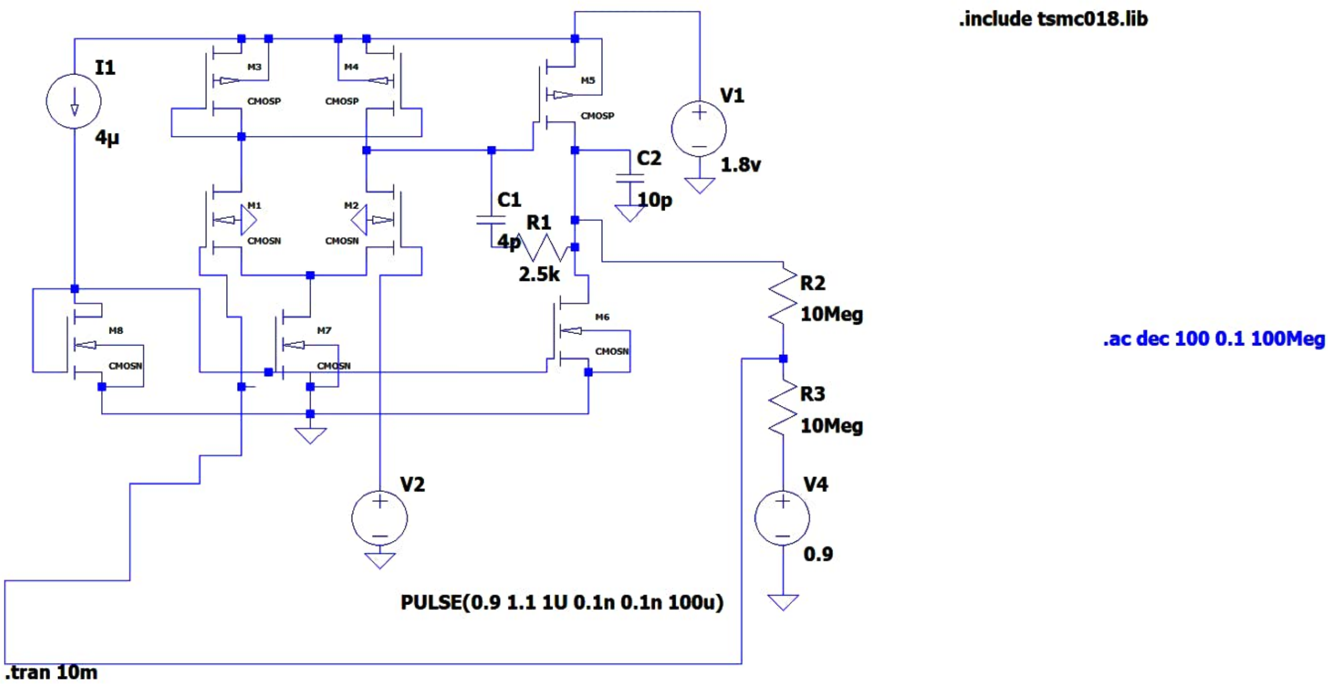
Freq:	4.7962176MHz	Mag:	-252.70729mdB
		Phase:	-116.34628°
		Group Delay:	12.406939ns

Cursor 2

Freq:	-- N/A--	Mag:	-- N/A--
		Phase:	-- N/A--
		Group Delay:	-- N/A--

Ratio (Cursor2 / Cursor1)

Freq:	-- N/A--	Mag:	-- N/A--
		Phase:	-- N/A--
		Group Delay:	-- N/A--


.include tsmc018.lib
I1
4µ
M3
M4
M5
CMOSP
CMOSP
CMOSP
V1
1.8v
C2
10p
C1
4p
R1
2.5k
M1
M2
CMOSN
CMOSN
M8
CMOSN
M7
CMOSN
M6
CMOSN
R2
10Meg
R3
10Meg
.ac dec 100 0.1 100Meg
V2
V4
0.9
PULSE(0.9 1.1 1U 0.1n 0.1n 100u)
.tran 10m

Operating Points

V(n001):	1.8	voltage
V(n002):	1.15744	voltage
V(n003):	1.19874	voltage
V(n004):	0.902604	voltage
V(n005):	0.9	voltage
V(n006):	0.902604	voltage
V(n007):	0.901302	voltage
V(n008):	0.0849597	voltage
V(n009):	0.557781	voltage
V(n010):	0.9	voltage
I(C1):	1.18453e-24	device_current
I(C2):	9.02604e-24	device_current
I(I1):	4e-06	device_current
I(R1):	0	device_current
I(R2):	1.30188e-10	device_current
I(R3):	1.30188e-10	device_current
I(V1):	-0.000106646	device_current
I(V2):	0	device_current
I(V4):	1.30188e-10	device_current
Ib(M1):	-1.26203e-12	device_current
Ib(M2):	-1.30332e-12	device_current
Ib(M3):	6.52559e-13	device_current
Ib(M4):	6.11263e-13	device_current
Ib(M5):	9.07396e-13	device_current
Ib(M6):	-9.12604e-13	device_current
Ib(M7):	-9.45853e-14	device_current
Ib(M8):	-5.67781e-13	device_current

```
Ib(M8):          -5.67781e-13   device_current
Id(M1):          2.79511e-05    device_current
Id(M2):          2.78263e-05    device_current
Id(M3):          2.79511e-05    device_current
Id(M4):          2.78263e-05    device_current
Id(M5):          4.68687e-05    device_current
Id(M6):          4.68686e-05    device_current
Id(M7):          5.57773e-05    device_current
Id(M8):          4e-06          device_current
Ig(M1):          0              device_current
Ig(M2):          0              device_current
Ig(M3):          -0             device_current
Ig(M4):          -0             device_current
Ig(M5):          -0             device_current
Ig(M6):          0              device_current
Ig(M7):          0              device_current
Ig(M8):          0              device_current
Is(M1):          -2.79511e-05   device_current
Is(M2):          -2.78263e-05   device_current
Is(M3):          -2.79511e-05   device_current
Is(M4):          -2.78263e-05   device_current
Is(M5):          -4.68687e-05   device_current
Is(M6):          -4.68686e-05   device_current
Is(M7):          -5.57773e-05   device_current
Is(M8):          -4e-06         device_current
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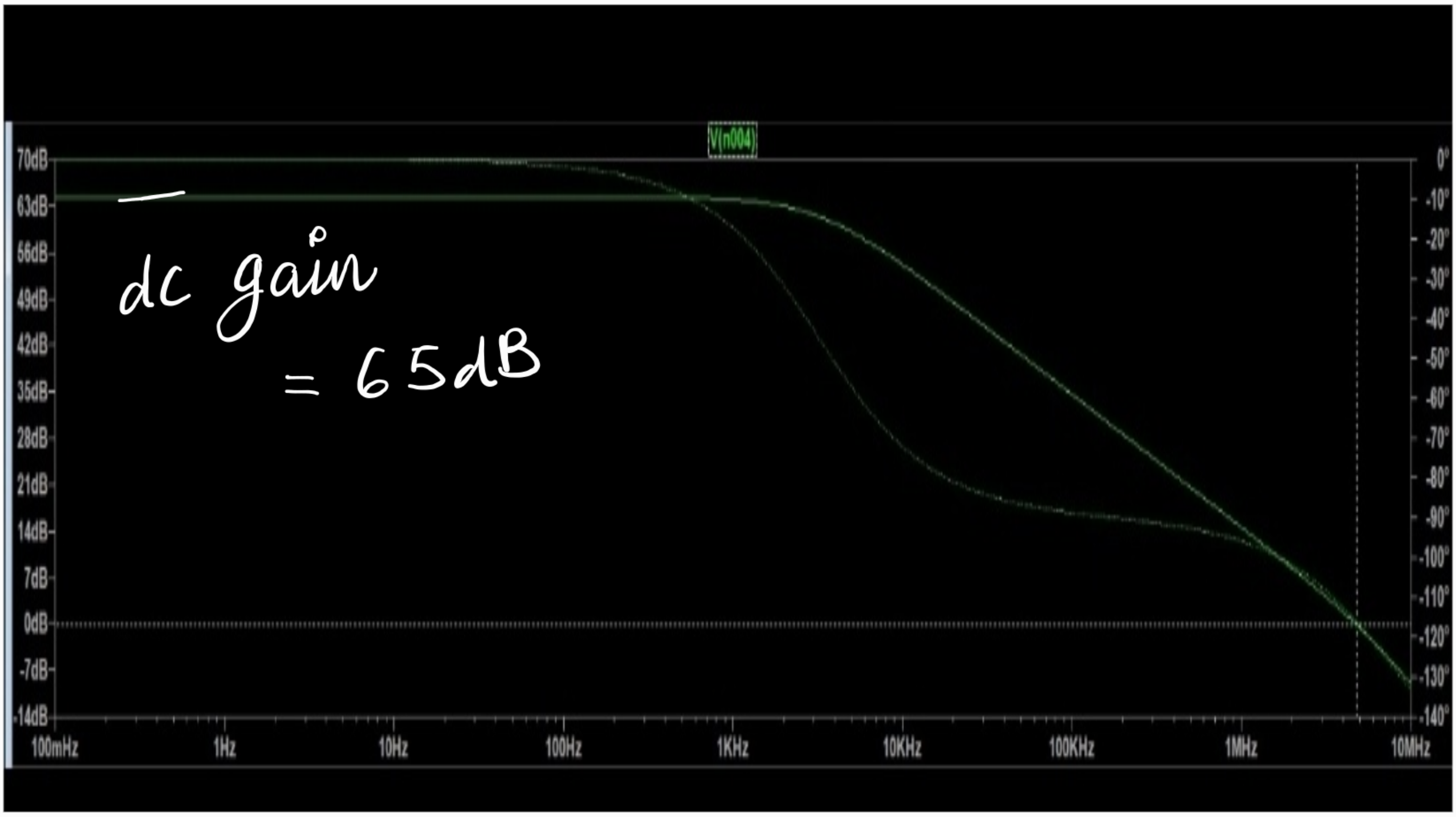
ota2ndstageprojectdipyaman240102116
V(n004)
1.38V
1.32V
1.26V
1.20V
1.14V
1.08V
1.02V
0.96V
0.90V
0.84V
0.78V
0µs
2µs
4µs
6µs
8µs
10µs
12µs
14µs
16µs
Gain = 2


V(n004)
70dB
63dB
56dB
49dB
42dB
35dB
28dB
21dB
14dB
7dB
0dB
-7dB
-14dB
0°
-10°
-20°
-30°
-40°
-50°
-60°
-70°
-80°
-90°
-100°
-110°
-120°
-130°
-140°
100mHz
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz
dc gain
= 65dB